The Mathematics Behind the Rubik's Cube

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1 Introduction

Have you ever played with a Rubik's Cube or seen your friends try to solve one? A Rubik's Cube is a 3x3x3 cube with six colors (red, blue, yellow, green, white, and orange). A pivot mechanism inside the cube allows you to twist each face so that the colors can be switched around. You try to solve the cube by getting the colors back together on the same corresponding side. Everyone who uses one usually tries to solve it as fast as possible, yet some fail while others can even do it in under 30 seconds! With all the different colors, they are not only eye-catching cubes of fun but also a research interest. Many want to learn the science and math behind this famous puzzle, but little do they know it could be so intriguing.

2 History

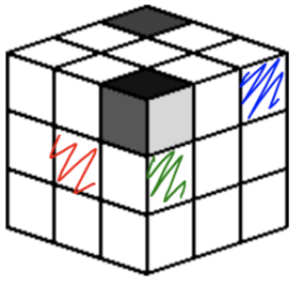
"We turn the cube and it twists us," said Erno Rubik, engineer of this unique design, as he explained the purpose of his creation. Rubik was a socialist bureaucrat who lived in Budapest, Hungary. He built the simple toy in his mother's apartment, not knowing that this invention would have an incredible impact on the future. He was first inspired in spring of 1974 by the popular 15 Puzzle. Invented in the late 1870s, this puzzle consists of 15 consecutively numbered, flat squares that can be slid around inside a square frame. A few years later, Sam Loyd, a recreational mathematician inspired by Erno Rubik, created a two-dimensional Magic Cube. It was later renamed to honor Erno Rubik to the Rubik's Cube. Since then, there have been many different cube variations made, but the one we will be discussing is called the standard 3x3x3.

3 Notation

Behind the 54 colorful little squares is a world of math that determine their possible arrangements, along with the countless ways to solve it. The basic notation for the six faces of the Rubik's Cube is F (front side), B (back), R (right), L (left), U (up), and D (down). Face rotations are represented by F for 90 degrees clockwise, F' for 90 degrees counterclockwise, and F^2 for a 180-degree rotation. To refer to individual 1x1 cubes within the Rubik's Cube, we use one letter to denote the center cube, two letters for an edge cube, and three letters for the corner cube. As shown in the following graphic, we refer to the red cube as at R (right), the green as at DL (down side, left), and the blue cube as at DUR (down side, up, right). The first letter always represents the side (face) of the Rubik's Cube where it is located.

4 Counting the Possibilities

When you try to solve a Rubik's cube, your way will probably be different from the way someone else tries to solve it. Why is that? It is because there are billions of ways to solve it due to the countless possibilities of moves. First, there are eight corner pieces that can be arranged in 8! $(8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1)$ or



40,320 different ways. Each has three orientations (because of the three colors on every corner piece), so there are 3^8 or 6,561 possibilities for each permutation of a corner piece. Then, you have 12 edge pieces with 2 orientations, resulting in 2^{12} or 4,096 permutations. However, there are some restrictions since in the solution for the Rubik's Cube, only $\frac{1}{3}$ of the permutations of the corner cubes are correct, and only $\frac{1}{2}$ of the permutations of the same-edge cubes are correct so that you are brought closer to the final result. The remaining 12 pieces of the cube (27 total pieces - 6 center pieces on the faces - 1 piece in the center of the cube that isn't visible - 8 corner pieces = 12) can be arranged in 12! or 4,790,001,600 ways.

Also, there is a $\frac{1}{2}$ correct cube rearrangement parity which results in the following equation which represents the total number of permutations in the Rubik's Cube:

$$\frac{8! \cdot 3^8 \cdot 12! \cdot 2^{12}}{3 \cdot 2 \cdot 2} = 4.3252 \cdot 10^{19}$$

However, after you make a move in the Rubik's Cube, you may also want to go back if you made a mistake. We also have to count those moves to find the precise number of arrangements. The arrangement involves a maximum and minimum number of steps. It is represented by the following equation, where $n \geq 19$:

$$12 \cdot 11^{n-1} \geq 4.3252 \cdot 10^{19}$$

In addition, we can represent cube permutations as a group of elements which we will call R. A group (G) consists of set of objects and a binary operator (*) in which it satisfies these four conditions:

1. The operation * is closed, so for any group elements h and g in G, $h * g$ is also in G.
2. The operation * is associative, so for any elements f, g, and h, $(f * g) * h = f * (g * h)$.
3. There is an identity element $e \in G$ such that $e * g = g * e = g$.
4. Every element in G has an inverse g^{-1} relative to the operation * such that $g * g^{-1} = g^{-1} * g = e$.

As a result, our binary operator for the Rubik's Cube represents all the cube moves, sequences or basically, the rotations of a face of the cube. This operation is closed since any face rotation gives us another permutation of the cube also in R. Rotations are also associative: it does not matter how we group them, as long as the order in which operations are performed is conserved. The inverse of a group element g is usually written as g^{-1}.

5 Basic Theorems

Like most mathematics, the Rubik's Cube has its own set of theorems:

1. The cube always has even parity, or an even number of cubies (small cubes within the large cube) exchanged from the starting position.

2. If the cube starts at the solved state, and one move sequence P is performed successively, then eventually the cube will return to its solved state.

These theorems help mathematicians further understand the Rubik's Cube as a whole, along with the math that describes the way it functions. Some of the common properties used to analyze the Rubik's Cube are associative and commutative properties. For the associative property, the permutations in the row can be grouped together. An example could be $(RY')L = R(Y'L)$. For the commutative property, although communativity is not a necessary condition of the permutation group, it shows when permutations are similar to each other, as in $FB = BF$. In addition, there is a neutral element which means there is a permutation that doesn't rearrange the set, like RR', and an inverse element which means every permutation has an inverse permutation, like $R - R'$.

6 Research

There are many researchers who examined and tested the Rubik's Cube to find the science/math behind it. Erik Demaine, a professor of computer science and engineering at MIT, explained, "The standard way to solve a Rubik's Cube is to find a square that's out of position and move it into the right place while leaving the rest of the cube as little changed as possible. That approach will indeed yield a worst-case solution that's proportional to $N2$." He and his colleagues had come to the conclusion that there exists a single sequence of twists that could move multiple squares into their proper places, cutting down the total number of moves. The only challenge that really stood in their way was how to describe those circumstances in mathematics, rather than words.

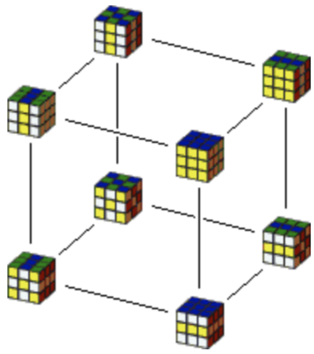
7 Lagrange's Theorem and Cayley Graph

However, an essential theorem used for the Rubik's Cube is Lagrange's Theorem, which states that if H is a subgroup of a finite group G, then the order of H divides the order of $G(|G|)$. This theorem was first developed by Joseph-Louis Lagrange in 1771. Some corollaries that follow from this theorem are:

1. Let G be a group and $g \in G$. Then $|g|$ divides $|G|$.

2. Let G be a group of prime order. Then G has no subgroups and hence is cyclic.

On the other hand, since a picture is worth a thousand words, a great way to gain insight into the structure of groups and subgroups of the Rubik's Cube is to draw a Cayley Graph, as shown below. The graph is unique and contains many properties. When graphing group G, each $g \in G$ is a vertex, each group generator $s \in S$ is assigned a color c_s, and lastly, for any $g \in G$, $s \in S$ elements corresponding to g and gs are joined by a directed edge of color c_s. Drawing a Cayley Graph for all real numbers would be infeasible since it involves about 43 trillion vertices. Instead, we should graph small subgroups of R.



8 Conclusion

In conclusion, analysis of the Rubik's Cube shows how math is a part of our everyday lives, not just our classes! Most of us only look at the Cube as a way to pass time or as just another fun activity to do with friends and family. Yet we rarely think about all the intriguing math that underlies it. Much more can be written about how math relates to this fascinating little puzzle. Keep looking for applications of math in all aspects of life!

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